# Logical equivalence identities ififf

### (Some) logical equivalences

$p \to q \equiv \neg p \vee q$

$p \to q \equiv \neg q \to \neg p$ **Contrapositive**

$\neg(p \to q) \equiv p \wedge \neg q$

$\neg(p \leftrightarrow q) \equiv p \oplus q$

$p \leftrightarrow q \equiv q \leftrightarrow p$

*Extra examples*:

$p \leftrightarrow q$ is not logically equivalent to $p \wedge q$ because _____

$p \to q$ is not logically equivalent to $q \to p$ because _____

# Logical operators truth tables

Truth tables: Input-output tables where we use $T$ for 1 and $F$ for 0.

| Input | | Output | | |
|---|---|---|---|---|
| | | **Conjunction** | **Exclusive or** | **Disjunction** |
| $p$ | $q$ | $p \wedge q$ | $p \oplus q$ | $p \vee q$ |
| $T$ | $T$ | $T$ | $F$ | $T$ |
| $T$ | $F$ | $F$ | $T$ | $T$ |
| $F$ | $T$ | $F$ | $T$ | $T$ |
| $F$ | $F$ | $F$ | $F$ | $F$ |
| | |  |  |  |

| Input | Output |
|---|---|
| | **Negation** |
| $p$ | $\neg p$ |
| $T$ | $F$ |
| $F$ | $T$ |
| |  |

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# Logical equivalence

**Logical equivalence** : Two compound propositions are **logically equivalent** means that they have the same truth values for all settings of truth values to their propositional variables.

**Tautology**: A compound proposition that evaluates to true for all settings of truth values to its propositional variables; it is abbreviated $T$.

**Contradiction**: A compound proposition that evaluates to false for all settings of truth values to its propositional variables; it is abbreviated $F$.

**Contingency**: A compound proposition that is neither a tautology nor a contradiction.

# Logical equivalence extra example

*Extra Example*: Which of the compound propositions in the table below are logically equivalent?

| Input | | Output | | | | |
|---|---|---|---|---|---|---|
| $p$ | $q$ | $\neg(p \wedge \neg q)$ | $\neg(\neg p \vee \neg q)$ | $(\neg p \vee q)$ | $(\neg q \vee \neg p)$ | $(p \wedge q)$ |
| $T$ | $T$ | | | | | |
| $T$ | $F$ | | | | | |
| $F$ | $T$ | | | | | |
| $F$ | $F$ | | | | | |

# Logical equivalence identities andornotxor

### (Some) logical equivalences

*Can replace $p$ and $q$ with any compound proposition*

$\neg(\neg p) \equiv p$ **Double negation**

$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$ **Commutativity** Ordering of terms

$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ **Associativity** Grouping of terms

$p \wedge F \equiv F$ $p \vee T \equiv T$ $p \wedge T \equiv p$ $p \vee F \equiv p$ **Domination** aka short circuit evaluation

$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$ **DeMorgan's Laws**

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